

SMART INDIA HACKATHON 2024



TITLE PAGE

- Problem Statement ID – **1647**
- Problem Statement Title- Development of **AI-ML based models for predicting prices** of **agri-horticultural commodities** such as pulses and vegetable (onion, potato, onion)
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID- **30095**
- Team Name (Registered on portal)- **Xebec's Crew**

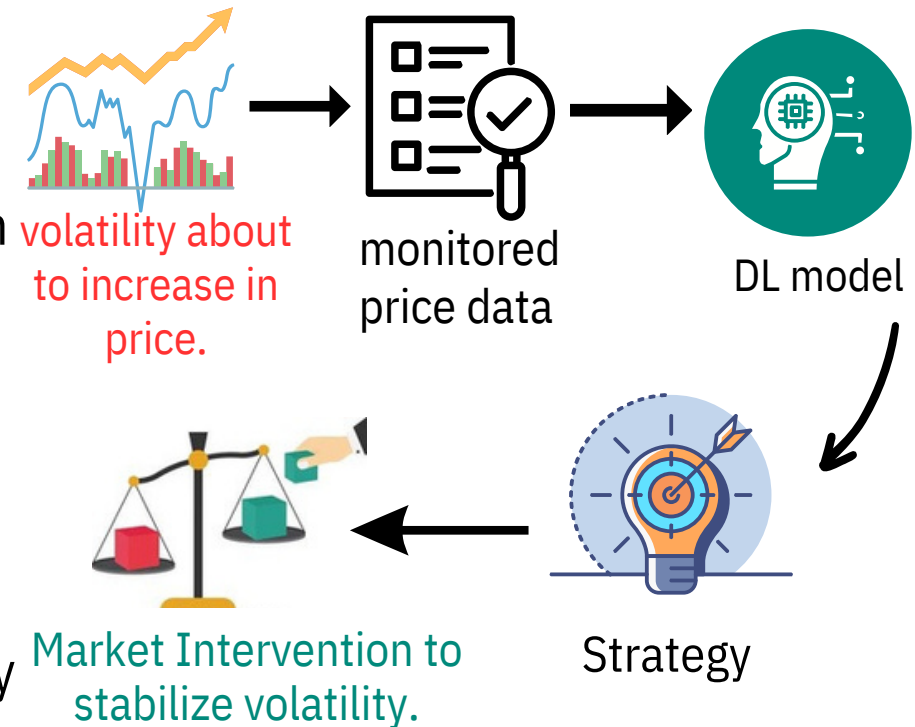
Quant Based Deep Learning Model

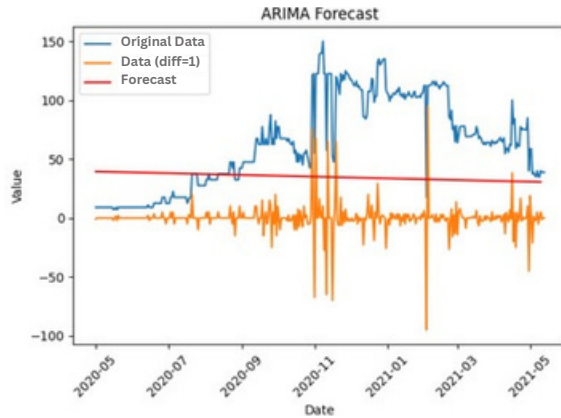
❖ Implementation of a Time-series based Deep Learning Model (LSTM+CNN Workflow) incorporated with quantitative financial analysis for strategic market interventions to stabilize the volatility in prices.

- The **DL spec TS model** captures the trend, seasonality and noise far better than ARIMA and SARIMAX, which gives the government a better control over the distribution of buffer stock.
- The **Bollinger Band** curve addresses the financial part of the problem i.e. where and when to send buffer stock and store stocks and pulses

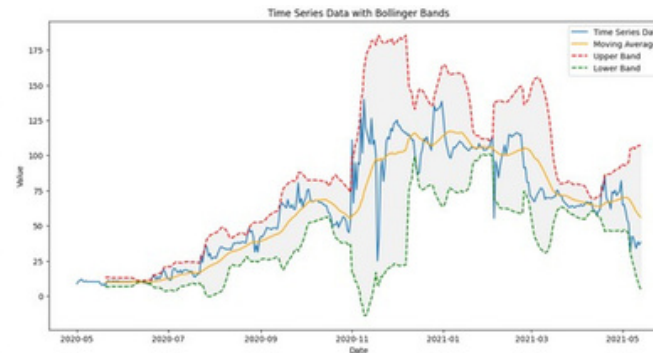
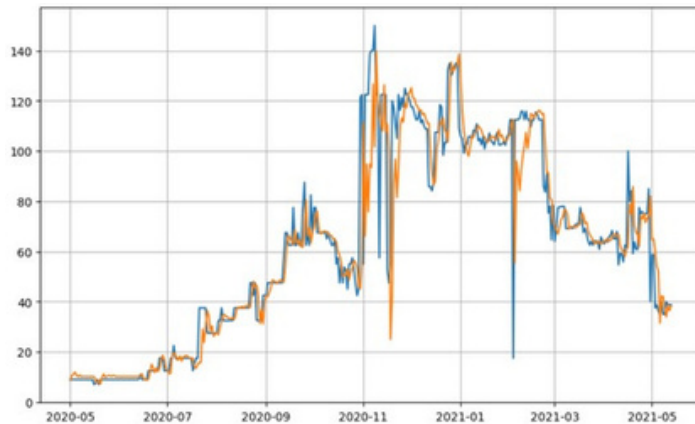
Innovation and uniqueness of the solution:

- Incorporation of **LSTM model**, which helps us to predict over long term data, **unlike ARIMA model**.
- Improvising a **CNN workflow** ensures the solidity of model as it take in account the uncertainty, short-term fluctuations, and random noises.
- **Deploying the whole model into a website** for convenience and easy access.





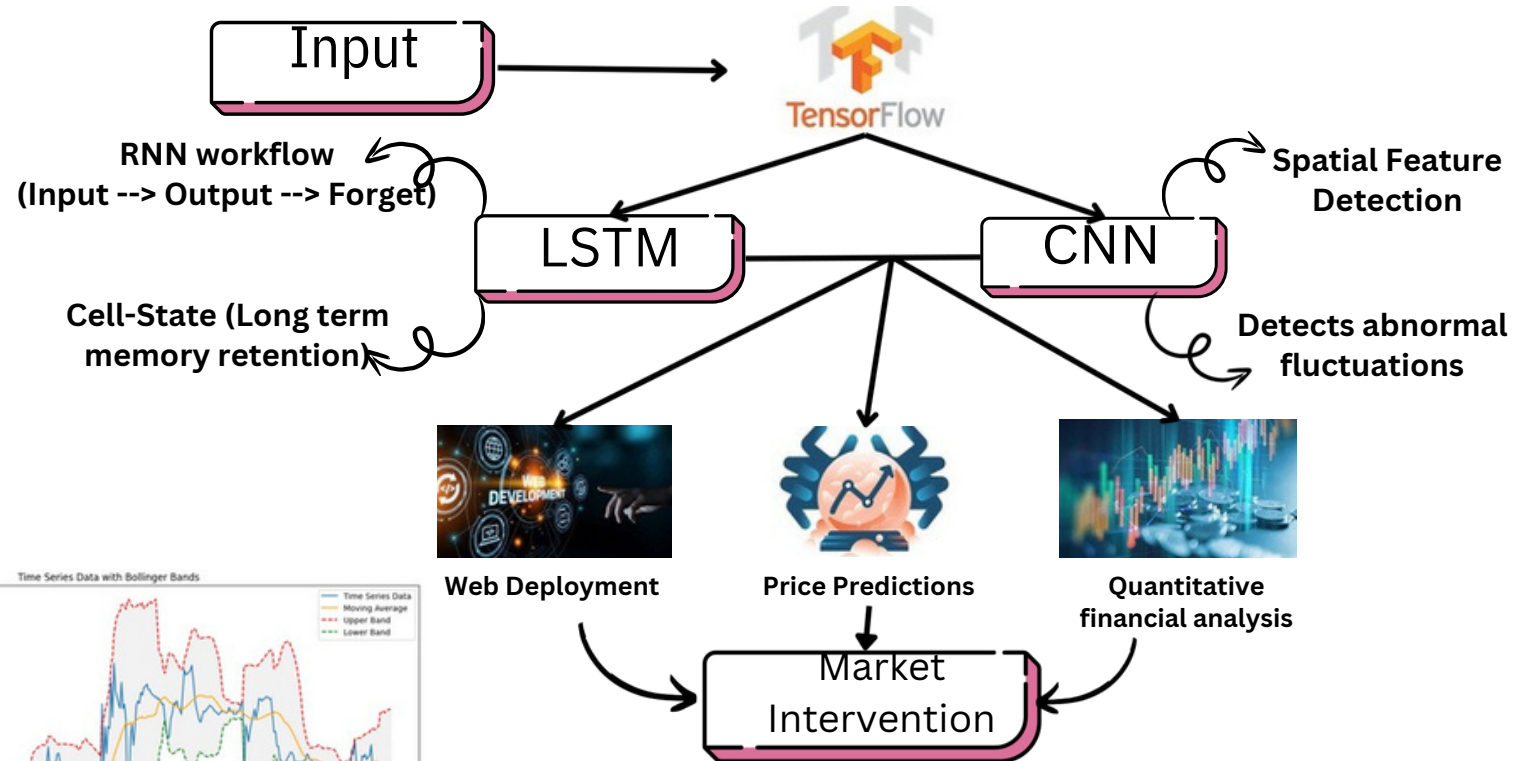
ARIMA Forecast



LSTM+CNN Model Forecast (screenshot taken from prototype model)

Bollinger Band Curve (specifying dates of interventions.)

prototype model: https://github.com/Falsegen1288/SIH_TS



Languages and Libraries Used:



FEASIBILITY AND VIABILITY

★ **High feasibility** as LSTM models have been proven **successful in finance, stock market prediction, and weather forecasting**—domains with similarities to agriculture. Major steps for completion are :-

- **Previously stored data.** (Crop Prices, weather forecast data, financial key metric data, stock buffer data along with location.)
- **Real-time data monitoring.**
- **Model implementation and testing.**
- **Market intervention.**

★ **Potential challenges and risks:** The biggest challenge is **real time data monitoring** and incorporating the data in model to make accurate predictions.

Another challenge can be to predict dates for interventions from the predicted price data.

★ **Strategies for overcoming these challenges:** Building an automated **AI model to append the real time data** in the input data set of our model to improve the predictions.

Solution of dates is covered in our model with dates that **overshoot the upper bollinger band** reported as dates for interventions.

IMPACT AND BENEFITS

IMPACT: Using the designed model for **successful market interventions** will certainly **stabilize the market price** in regions with a depreciation in crop price due to decrease of the crop in that region, ultimately leading to **availability of the crop at affordable price.**

Economic Benefit:

- Price stability.
- Increased farmer income by securing fair prices.
- Cost efficiency.
- Improved market predictability for stakeholders.

Social Benefit:

- Farmer livelihood security.
- Enhanced food security.
- Reduced poverty.
- Less social unrest caused by food price spikes.

Environmental Benefit:

- Sustainable farming practices due to better planning and forecasting.
- Reduced food waste by balancing supply and demand.

RESEARCH AND REFERENCES

(Details / Links of the reference and research work)



1. Time series analysis article

2. Article on Deep Learning based LSTM (Long - short term memory) model

3. Article on how to combine a LSTM & CNN framework

4. Research paper on Bollinger Band Curve